BMS PROJECT 1

Systematic Review: Modeling Approaches for Antibody Afucosylation Mechanistic · Machine Learning · Hybrid Models · AI

Presented By:

TEAM DATA MINDS

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Problem Statement: What Are We Solving?

Fucosylation = the attachment of core fucose on **N-linked glycans → Critical Quality Attribute** (CQA) in **mAb** biologics manufacturing → Directly impacts **ADCC efficacy**, patient safety, and regulatory approval.

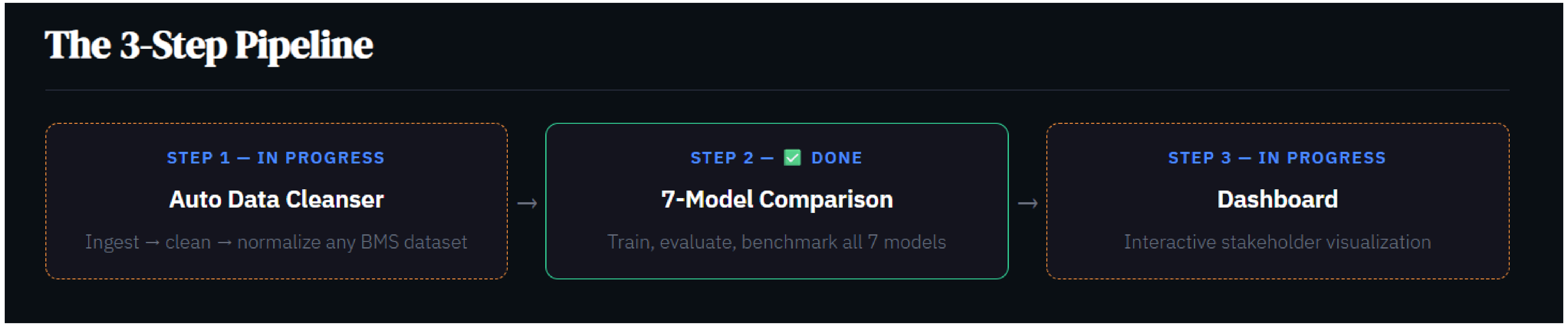
The Challenge:

- Current prediction methods are **slow**, costly, or lab-scale only.
- No scalable industrial solution exists.
- Process variables interact in complex, nonlinear ways, linear methods fail to capture this.

The Opportunity — ML & AI:

- Predict afucosylation in real time or early in the process using routinely measured process variables.
- Replace slow analytical assays with fast, data-driven surrogate models.
- Enable proactive batch correction before CQA goes out of specification.
- Build interpretable models that satisfy regulatory (FDA Quality by Design) requirements.
- Generalize across multiple mAb products using a single unified modeling framework.

7 ML/AI models benchmarked on a mechanistically-grounded synthetic dataset — evaluated on accuracy, interpretability, data requirements, and regulatory usability.



Dataset Overview: Synthetic Dataset

Why Synthetic?

- Real BMS CHO bioprocess data is proprietary & unpublished
- Built a mechanistically-grounded synthetic dataset instead
- Every variable & coefficient is anchored in published biochemistry
- Validates the pipeline, not just the predictions — ground truth is known

Whats in it?

- 10 input variables: GDP-Fucose, Mn^{2+}, Uridine, pH, DO, pCO_2, Temperature, VCD, Osmolality, Lactate
- Target: Fucosylation % (0–100%)
- Nonlinear ground truth: Michaelis–Menten saturation, quadratic pH & temperature optima, GDP-Fucose × Mn^{2+} interaction
- Batch effects + heteroscedastic noise layered on top

Two Dataset Sizes

- BMS1: **N = 500, & 10 batches**
- BMS2: **N = 10,000, & 50 batches**
- Only 3 parameters changed that isolates dataset size as the single experimental variable

Realism Layers Added

- **Missing data — MCAR**: 5% of values dropped completely at random across features
- **Missing data — MNAR**: DO sensor failure at high DO (>70%)
- **4 noise variants:** low / medium / high / extreme - (0.5× to 4× base noise)
- **Time-series variant:** 20 batches × 14 days

References: Raju et al. (2000) · Hossler et al. (2009) · Jedrzejewski et al. (2014)

Model 1: Ridge Regression

What it does:

- **L2-regularised** linear regression — fully interpretable coefficients
- **α** selected via **5-fold cross-validation** (RidgeCV)
- **Diagnostic checks**: residual normality, homoscedasticity (Breusch-Pagan), VIF, Cook's distance

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	0.513	0.573
RMSE	3.44	3.76
Top drivers	GDP-Fucose, Temperature, Mn^{2+}	Same — stable

Key takeaway: Linear ceiling at ~0.57 , cannot capture quadratic optima or interaction terms regardless of data size

Model 2: Partial Least Squares Regression (PLSR)

What it does:

- Projects inputs onto **latent components** that maximally covary with fucosylation
- Optimal components selected by 5-fold **CV → 4 components**
- **VIP scores** give clean feature importance ranking — gold standard in pharma PAT/QbD

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	0.510	0.573
Top VIP features	GDP-Fucose (1.92), Temp (1.76), Mn^{2+} (1.20)	Same top 3, pH drops to VIP = 0.02

Key takeaway: Same predictive ceiling as Ridge, but VIP scores are regulator-friendly and directly submittable in a PAT filing

Model 3: Random Forest

What it does:

- **300 trees**, max depth 15, captures **nonlinear effects** and interactions naturally
- Two importance metrics: **MDI (biased) + Permutation importance** (unbiased) — cross-checked
- **OOB score** used as honest generalisation estimate

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	0.659	0.789
RMSE	2.88	2.64
Top features	Temperature, GDP-Fucose, pH	GDP-Fucose, Temperature, pH
OOB Score	0.716	0.778

Key takeaway: First model to break through the linear ceiling and tree splits naturally recover quadratic pH and temperature optima

Model 4: Artificial Neural Network (ANN)

What it does:

- **Architecture:** 128 → 64 → 32, ReLU, Adam — ~12,000 parameters
- Early stopping, adaptive learning rate, L2 regularisation
- **LIME** used for local post-hoc explanations

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	0.049 ❌	0.805 ✅
RMSE	4.80	2.54
Parameters vs samples	12,000 params / 400 samples	12,000 params / 8,000 samples

Key takeaway: +1,440% R^2 improvement — the ANN requires data. At N=500 it is the worst model. At N=10,000 it is among the best.

Model 5: Gaussian Process Regression (GPR)

What it does:

- **Probabilistic model** — returns a prediction and a calibrated uncertainty bound
- **RBF kernel + WhiteKernel** noise term, optimised via 3 restarts
- 4 kernels compared: RBF wins
- Note: **$O(n^3)$ cost → capped** at **300-sample training** subsample in both experiments

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	**0.749** — best on BMS1	0.793
95% CI Coverage	95.0% ✅	95.95% ✅
Mean σ	2.34%	2.69%

Key takeaway: Best small-data model — calibrated uncertainty is its unique value. $O(n^3)$ cost limits scaling.

Model 6: Hybrid Physics-Informed

What it does:

- **5 engineered features** built from known biochemical mechanisms:
  - **FPI** — GDP-Fucose × Mn^{2+} (FUT8 substrate × cofactor)
  - **Energy Charge** — DO, Lactate, pCO_2 (metabolic balance)
  - **Golgi Proxy** — Uridine, pH, Temperature
  - **Stress Index** — Osmolality, Lactate, pCO_2
  - **Enzyme Activity** — Mn^{2+}, pH, Temperature
- **3 RF variants** compared: raw only / engineered only / hybrid (raw + engineered)

Variant	BMS1 R^2	BMS2 R^2
Raw (10 features)	0.688	0.824
Engineered only (5 features)	0.467	0.616
Hybrid (15 features)	0.722	0.824

Key takeaway: 5 engineered features retain 64.7% (BMS1) → 74.7% (BMS2) of full model performance and half the variables, three-quarters of the signal

Model 7: XGBoost + SHAP

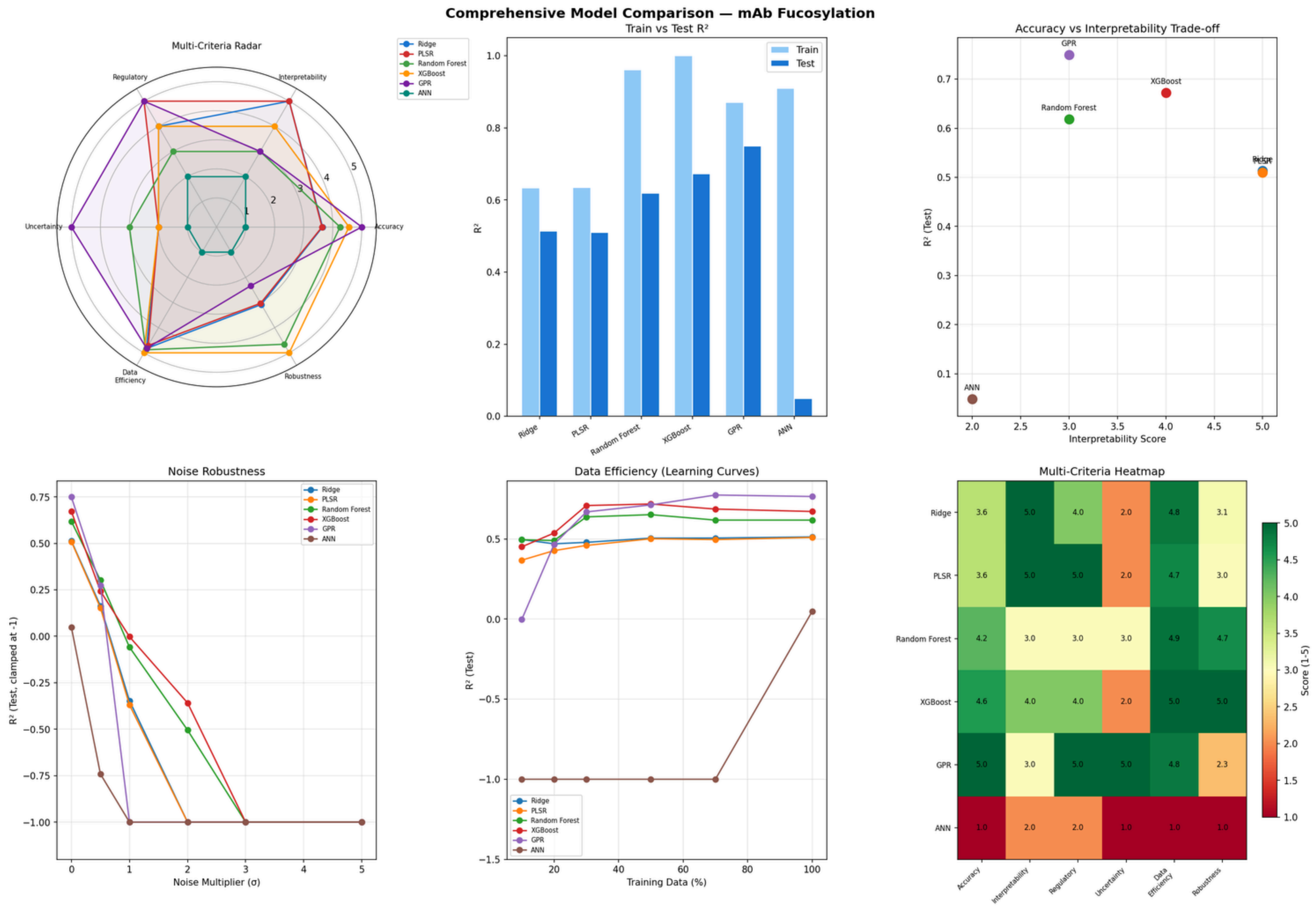
What it does:

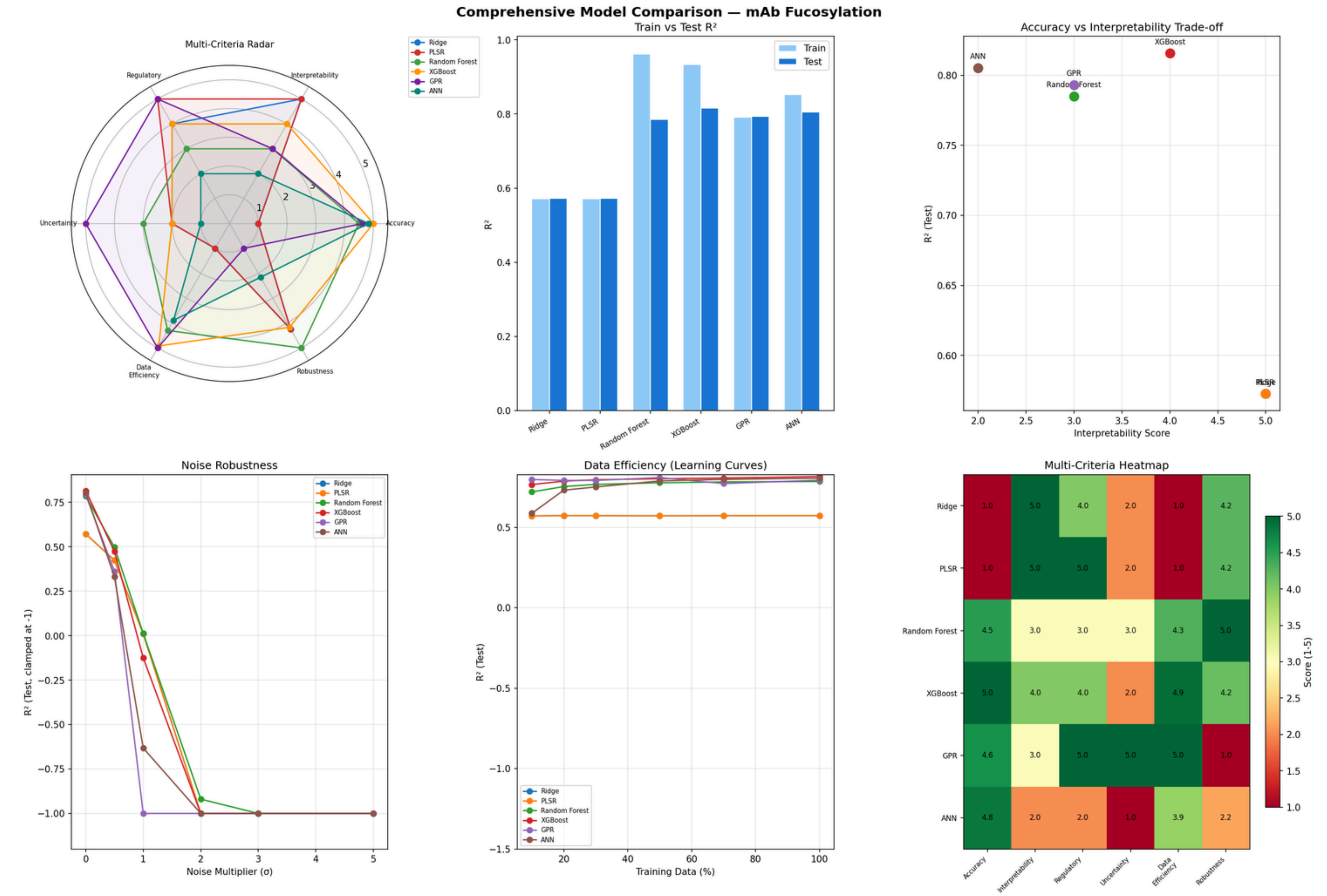
- **Gradient boosted trees**: 300 estimators, max depth 5, learning rate 0.1
- **SHAP (TreeSHAP)** — theoretically grounded local + global explanations
- **Three SHAP analyses:** global ranking, summary plot, pairwise interaction matrix

	BMS1 (N=500)	BMS2 (N=10,000)
R^2 Test	0.743	**0.827**
RMSE	2.50	2.39
Top SHAP features	Temperature, GDP-Fucose, pH, Mn^{2+}	Same order — stable

Key takeaway: Best production model with highest accuracy at N=10,000 + full SHAP explainability. SHAP rankings match the ground truth.

Cross-Model Comparison: Performance, Noise & Data Efficiency





Key takeaway: 1) Linear models hit a hard ceiling (~0.57) — not a data problem, a structure problem

2) ANN: worst model at N=500 → tied for best at N=10,000

Cross-Model Comparison: Multi-Criteria Rubric & Data Efficiency

Model	R^2 BMS1	R^2 BMS2	Δ
Ridge	0.513	0.573	+0.06
PLSR	0.510	0.573	+0.06
Random Forest	0.618	0.785	+0.17
XGBoost	0.672	0.816	+0.14
GPR	0.749	0.793	+0.04
ANN	**0.049**	**0.805**	**+0.76**

Model	Accuracy	Interpretability	Regulatory	Uncertainty	Total (BMS1)	Total (BMS2)
GPR	5.0	3.0	5.0	5.0	**25.2**	23.6
XGBoost	4.6	4.0	4.0	2.0	24.6	**24.1**
PLSR	3.6	5.0	5.0	2.0	23.4	18.2
ANN	1.0	2.0	2.0	1.0	8.0	15.9

AI-Powered Dynamic ML Pipeline for Fucosylation Prediction

Feature	Description
📂 Data Upload	Upload any BMS or custom CHO bioprocess dataset (CSV)
🧹 Auto Data Cleansing	One-click missing value imputation, outlier detection (IQR, Z-score, Isolation Forest), and batch drift flagging via KS test
🔍 Feature Selection	SHAP-based, VIP score, or MDI ranking of input variables across all 7 models
🧠 Model Selection & Training	Dropdown to choose from all 7 models (Ridge, PLSR, RF, ANN, GPR, Hybrid, XGBoost); trains on uploaded data in real time
📈 Results Dashboard	R^2 metrics, actual vs. predicted scatter plots, learning curves, and multi-criteria radar chart per model
🤖 AI Report Generation	**Gemini 2.5 Flash** auto-generates a regulatory-style report (ICH Q8/Q9 aligned) in three formats: Executive Summary, Technical Deep-Dive, Regulatory Submission
📊 BMS1 vs BMS2 Reference	Static side-by-side comparison of small vs. large dataset model performance, always accessible
📄 Report Export	Download PDF/Word regulatory summary with model outputs, SHAP attributions, and uncertainty bounds

Automated Dataset Cleansing Module (Stage 1 Pipeline)

Step	Method	What It Catches
📥 Data Ingestion	Upload CSV; schema auto-detected and validated	Column mismatches, wrong data types, empty files
🔎 Missing Value Detection	Flags MCAR vs. MNAR patterns; imputes via median/KNN	DO probe failures (>70%), pCO_2 drift (>100 mmHg), random gaps
📉 Outlier Detection	Three simultaneous methods: IQR, Z-score, Isolation Forest	Sensor spikes, erroneous readings, batch contamination events
🔄 Batch Drift Detection	Kolmogorov–Smirnov (KS) test across batch IDs	Lot-to-lot variation, equipment drift, operator shifts
⚖️ Feature Scaling	Min-max normalization to [0,1] per biological operating range	Ensures all 10 variables are on comparable scales before modeling
✅ Schema Enforcement	Output normalized to fixed 10-feature schema expected by model bench	Guarantees plug-and-play compatibility with all 7 downstream models
📋 Cleanse Report	Summary of imputed values, flagged outliers, and drift alerts	Full audit trail for regulatory documentation (ICH Q9 alignment)

Platform Access & User Support Features

Feature	Description
🔐 Login	**Passwordless SaaS login via magic link** — enter email, receive a one-time link (10-min expiry), delivered via Gmail SMTP; no password stored
👤 My Profile	View account details, usage history, previously uploaded datasets, and saved model runs
ℹ️ About & FAQ	Overview of the pipeline, model descriptions, and answers to common questions on model selection and output interpretation
🎧 Support System	In-app support ticket submission, contact form, and documentation links for troubleshooting uploads, model errors, and report generation